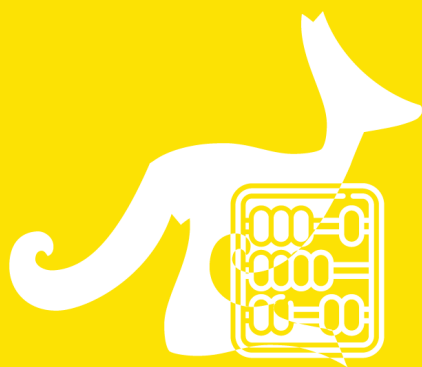




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

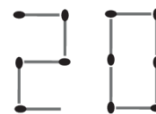
Good Luck!

预祝考试顺利

**Year 9
&
Year 10**

3 points

1. Carola is forming the four-digit number 2022 using some matches from a box. The box originally contained 30 matches. She has already started and formed the first two digits, as shown in the diagram. How many matches will remain in the box when she has finished forming 2022?



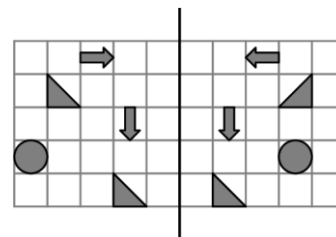
Carola 用火柴盒里的火柴来摆成四位数字 2022。火柴盒原先有 30 根火柴，她已经摆好了前两位数字，如图所示。请问当她摆好 2022 后，火柴盒里还剩多少根火柴？

- (A) 20 (B) 19 (C) 10 (D) 9 (E) 5

2. An equilateral triangle of side 12 has the same perimeter as a square of side x . What is the value of x ?
一个边长为 12 的等边三角形与一个边长为 x 的正方形的周长相同，请问 x 的值是多少？

- (A) 9 (B) 12 (C) 16 (D) 24 (E) 36

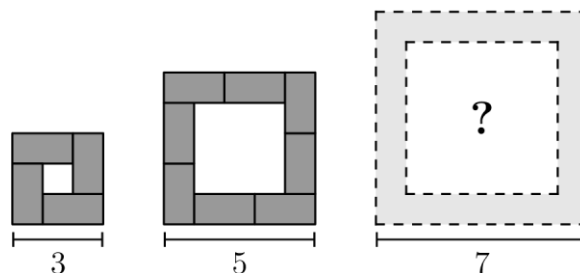
3. Some shapes are drawn on a piece of paper. The teacher folded the left-hand side of the paper over the thick line. How many of the shapes on the left-hand side will fit exactly on top of a shape on the right-hand side?



如图所示，一张纸上画有一些图形。老师将纸的左边沿着粗线对折。请问纸张左边有多少个图形恰好和右边的图形重合？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

4. Katrin arranges tables of size 2×1 according to the number of participants in a meeting. The diagrams show a top view of the tables for a small, a medium and a large meeting. How many tables are used for the large meeting?



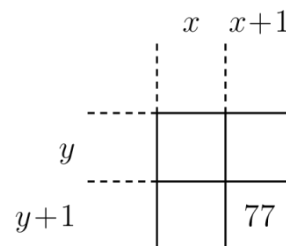
Katrin 根据参会者数量准备了一些大小为 2×1

的小桌子，右图分别表示小型、中型和大型会议桌子摆放的俯视图。请问大型会议一共需要用到多少张小桌子？

- (A) 10 (B) 11 (C) 12 (D) 14 (E) 16

5. A square of numbers is taken out from a multiplication table. Only one number is visible. The integers x and y are both positive and x is greater than y . What is the value of x ?

从乘法表中取出一个小正方形，其中有四个数，但只有一个数字是已知的。 x 和 y 都是正整数且 x 大于 y 。请问 x 的值是多少？



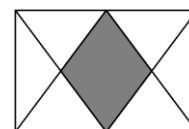
- (A) 6 (B) 7 (C) 8 (D) 10 (E) 11

6. I am less than my half and greater than my double. The sum of me and my square is zero. Who am I?
一个数比它的一半小，但比它的两倍大，且该数与它的平方相加等于零，请问这个数是多少？

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

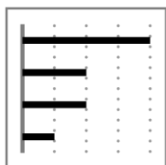
7. In the rectangle shown, the midpoints of the two longer sides are joined to all four vertices. What fraction of the rectangle is shaded?

如图所示，矩形较长两边的中点连接到其四个顶点。请问阴影部分占矩形总面积的几分之几？



- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{2}{7}$ (D) $\frac{1}{3}$ (E) $\frac{2}{5}$

8. On Nadya's smartphone, this diagram shows how much time she spent last week on each of her apps.
下图表示 Nadya 上周在每个手机应用程序上的使用时长。



This week she halved the time spent on two of these apps, but spent the same amount of time on the other two apps. Which of the following could be the diagram for this week?

这周，她花在两个应用上的时间仅为上周一半，但花在另外两个应用上的时间都与上周相同。请问下面哪张图能表示本周的使用时长？

- (A) (B) (C) (D) (E)

9. There are five candidates in the school election. After 90% of the votes had been counted, the preliminary results were as follows:

学生会选举中有五名候选人。在统计了 90% 的选票之后，初步结果如下：

Alex	Bella	Calvin	Diane	Eddy
14	11	10	8	2

How many students still have a chance of winning the election?

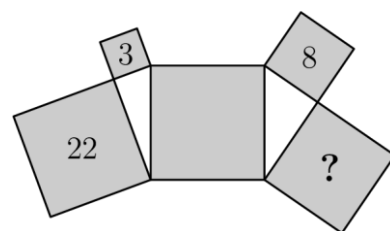
请问有多少名候选人有机会在选举中获胜？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

10. Five squares and two right-angled triangles are arranged as shown.

The numbers 3, 8 and 22 inside three of the squares indicate their areas in square metres. What is the area of the square containing the question mark?

五个正方形和两个直角三角形的排列方式如右图所示。其中三个正方形内的数字 3、8 和 22 表示其面积，单位为 m^2 。请问标记问号的正方形的面积是多少？

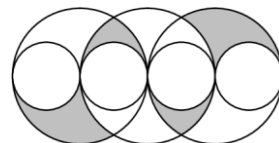


- (A) $14m^2$ (B) $15m^2$ (C) $16m^2$ (D) $17m^2$ (E) $18m^2$

4 points

11. The diagram shows three large circles of equal radius and four small circles of equal radius where the centers of all circles and all points of contact lie on one straight line. The radius of each small circle is 1. What is the shaded area?

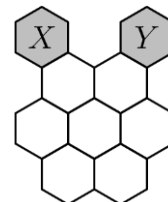
图中显示了三个半径相等的大圆和四个半径相等的小圆，其中所有圆心和所有切点都位于同一条直线上。已知每个小圆的半径为 1，请问阴影部分的面积是多少？



- (A) π (B) 2π (C) 3π (D) 4π (E) 6π

12. Apini moves from hexagon X to hexagon Y . She can only move from one hexagon to another if they have an edge in common. How many different routes are there from X to Y that pass through each of the seven white hexagons exactly once?

Apini 要从六边形 X 移动到六边形 Y 。只有两个六边形有一条公共边时，她才能从一个六边形移动到另一个六边形中。请问从 X 移动到 Y ，一共有多少条路线恰好可以经过所有七个白色六边形且都只经过一次？



- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

13. I once met six siblings whose ages were six consecutive whole numbers. I asked each of them the question: "How old is your oldest sibling?" Which of the following could **not** be the sum of their six answers?

我曾经遇见过六兄妹，他们的年龄恰好是六个连续的整数。然后我问他们每个人同一个问题：“你最大的兄弟/姐妹多大了？”请问以下哪个选项**不可能**是他们六个人答案的总和？

(A) 95 (B) 125 (C) 167 (D) 205 (E) 233

14. Eva puts 2022 tiles in a long line. Then Adam removes every sixth tile. Next Beata removes every fifth tile from those that remain. Then Calle removes every fourth tile. Finally, Doris removes all the remaining tiles. How many tiles does Doris remove?

Eva 将 2022 块瓷砖排成一长列。然后 Adam 每数到第六块瓷砖就把这块瓷砖拿走；全部拿走之后，Beata 开始数剩下的瓷砖，每数到第五块就把这块瓷砖拿走。随后，Calle 继续数剩下的瓷砖，每数到第四块就把那块瓷砖拿走。最后，Doris 拿走了所有剩余的瓷砖。请问 Doris 拿走了多少块瓷砖？

(A) 0 (B) 337 (C) 674 (D) 1011 (E) 1348

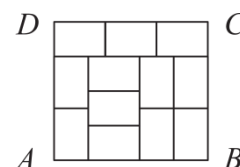
15. Three children asked their grandmother how old she was. She replied by asking them to guess her age. One child said she was 75, one said she was 78 and one said she was 81. It turned out that one of the guesses was wrong by 1 year, one was wrong by 2 years and one was wrong by 4 years. What is the grandmother's age?

三个小孩问他们祖母的岁数。祖母让他们猜猜她的年龄。一个小孩说 75 岁，一个说 78 岁，还有一个说 81 岁。实际上他们都猜错了，一个小孩差了 1 岁，一个小孩差了 2 岁，还有一个小孩差了 4 岁。请问祖母多大岁数了？

(A) 76 (B) 77 (C) 79 (D) 80 (E) cannot be determined exactly 无法确定

16. The diagram shows a large rectangle $ABCD$ divided into 12 identical small rectangles. What is the ratio AD/DC ?

如图所示，大矩形 $ABCD$ 被分成了 12 个完全相同的小矩形。请问 AD/DC 等于多少？



(A) $8/9$ (B) $5/6$ (C) $7/8$ (D) $2/3$ (E) $9/8$

17. A rabbit and a hedgehog had a race around a 550 m long circular track. Both ran at constant speeds. The rabbit's speed was 10 m/s, and the hedgehog's speed was 1 m/s. They started at the same time. However, the hedgehog ran in the opposite direction to the rabbit. When they met, the hedgehog immediately turned round and ran after the rabbit. How long after the rabbit did the hedgehog reach the finish?

一只兔子和一只刺猬在一条 550 米长的环形跑道上赛跑。它们都匀速奔跑，兔子的速度是 10 米/秒，刺猬的速度是 1 米/秒。它们同时出发，但刺猬跑的方向与兔子相反。当它们相遇时，刺猬立即掉头追赶兔子。请问兔子到达终点后多久，刺猬才到达终点？

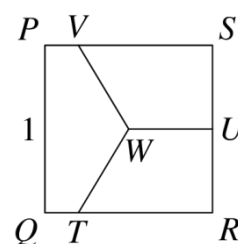
(A) 45 sec 45 秒 (B) 50 sec 50 秒 (C) 55 sec 55 秒 (D) 100 sec 100 秒 (E) 505 sec 505 秒

18. The diagram shows square $PQRS$ of side-length 1. The midpoint of RS is marked U and the centre of the square is marked W . Line segments TW , UW and VW split the square into three regions of equal area.

What is the length of SV ?

右图表示边长为 1 的正方形 $PQRS$ ，边 RS 的中点标记为 U ，正方形的中心标记为 W 。线段 TW 、 UW 和 VW 将正方形划分成三个面积相等的区域。

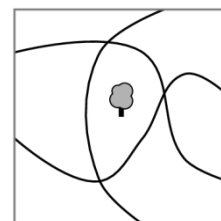
请问 SV 的长度是多少？



(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$ (E) $\frac{5}{6}$

19. There are three paths through our city park. A tree is planted in the middle of the park, as shown. What is the smallest number of trees that need to be planted so that there are the same number of trees on both sides of each of the paths?

有三条道路穿过城市公园。如图所示，公园中间种植有一棵树。请问至少还需要种植多少棵树，才能使得每条道路两侧的树木总数相同？



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

20. Veronica has five rings on her fingers, as shown in the diagram. She takes them off one at a time. In how many different ways can she do this?

如图所示，Veronica 的手指上戴着五枚戒指。她每次取下一枚戒指，直至从手上全部取下。请问她一共有多少种取法？

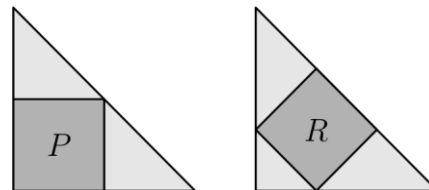


(A) 16 (B) 20 (C) 24 (D) 30 (E) 45

5 points

21. Two congruent isosceles right-angled triangles each have a square inscribed, as shown in the diagram. The square marked P has an area of 45. What is the area of the square marked R ?

两个全等的等腰直角三角形各有一个内接正方形，如图所示。标记 P 的正方形面积是 45。请问标记 R 的正方形面积是多少？



- (A) 35 (B) 40 (C) 45 (D) 50 (E) 60

22. Eight teams participate in a football tournament. Each team plays against each other team exactly once. In each match, the winner gets 3 points and the loser does not get any points. If a match is drawn, each team gets 1 point. At the end of the tournament the total number of points obtained by all the teams is 61. What is the largest number of points that the champion team could have obtained?

八支球队参加一场足球比赛。每支球队和其他所有球队各比赛恰好一次。在每场比赛中，胜方得 3 分，败方不得分。如果平局，每队都得 1 分。比赛结束时，所有球队的得分总和为 61 分。请问冠军队最多能得多少分？

- (A) 21 (B) 19 (C) 18 (D) 17 (E) 16

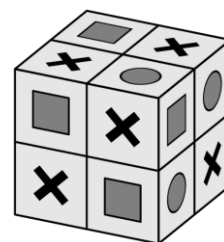
23. A group of pirates divided 200 gold coins and 600 silver coins between them. Each officer received 5 gold and 10 silver coins. Each sailor received 3 gold and 8 silver coins. Each cabin boy received 1 gold and 6 silver coins. How many pirates are there in the group?

一群海盗瓜分完 200 枚金币和 600 枚银币后，海盗中的每名长官得到 5 枚金币和 10 枚银币，每位水手得到 3 枚金币和 8 枚银币，每名男服务员得到 1 枚金币和 6 枚银币。请问一共有多少个海盗？

- (A) 50 (B) 60 (C) 72 (D) 80 (E) 90

24. The squares on the surface of a $2 \times 2 \times 2$ cube have one of three shapes on them. The shapes are either a circle or a square or an X sign. Any two squares that share a common side have different shapes on them. The picture shows one such possibility. Which of the following combinations of shapes is also possible on such a cube?

一个 $2 \times 2 \times 2$ 的立方体表面上的小正方形上标有三种图案中的一种。这些图案分别是圆形、正方形和 X 符号。任意有公共边的两个小正方形都标有不同的图案。如图所示表示一种可能的图案组合。请问这个立方体上也有可能出现下列哪种图案组合？



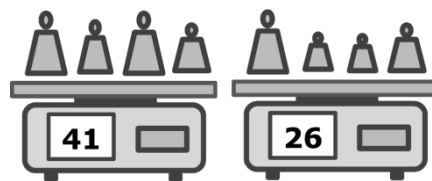
- (A) 6 circles, 8 squares and the rest are X's 6 个圆形，8 个正方形，其余都是 X 符号
(B) 7 circles, 8 squares and the rest are X's 7 个圆形，8 个正方形，其余都是 X 符号
(C) 5 circles, 8 squares and the rest are X's 5 个圆形，8 个正方形，其余都是 X 符号
(D) 7 circles, 7 squares and the rest are X's 7 个圆形，7 个正方形，其余都是 X 符号
(E) none of the previous 以上都不可能

25. The inhabitants of a city always speak by means of questions. There are two types of inhabitants: the “positives”, who always ask questions for which the answer is “yes” and the “negatives” who always ask questions for which the answer is “no”. I met Albert and Berta and Berta asked me “Are Albert and I both negative?”. What type of inhabitants are Albert and Berta?

某城市的居民总是通过提问的方式来交流。这里有两种类型的居民：一种是“积极的”，他们总是问答案为“是”的问题；另一种是“消极的”，他们总是问答案为“否”的问题。我遇见了 Albert 和 Berta，Berta 问我：“Albert 和我都是‘消极的’居民吗？”。请问 Albert 和 Berta 是哪种类型的居民？

- | | |
|---|----------------------------|
| (A) Both are positives | 两人都是“积极的” |
| (B) Both are negatives | 两人都是“消极的” |
| (C) Albert positive, Berta negative | Albert 是“积极的”，Berta 是“消极的” |
| (D) Albert negative, Berta positive | Albert 是“消极的”，Berta 是“积极的” |
| (E) There are not enough information to decide. | 信息不足，无法推断出来 |

26. A grocer has twelve different integer weights from 1 kg to 12 kg. She splits them into three groups of four weights each. The total weight of the first group is 41 kg and of the second is 26 kg. Which of the following weights is in the same group as the weight of 9 kg?

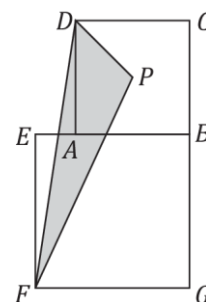


一个杂货商有 12 种不同重量的砝码，重量从 1 kg 到 12 kg 不等，且均为整数。她把这些砝码分成三组，每组有四个砝码。第一组的总重量为 41 kg，第二组的总重量为 26 kg。请问以下哪个重量的砝码与 9 kg 的砝码在同一组？

- (A) 3 kg (B) 5 kg (C) 7 kg (D) 8 kg (E) 10 kg

27. The lengths of the diagonals of the squares $ABCD$ and $EFGB$ are 7 cm and 10 cm respectively. The point P is the intersection of the diagonals of the square $ABCD$. What is the area of the triangle FPD ?

正方形 $ABCD$ 和 $EFGB$ 的对角线长度分别为 7 cm 和 10 cm。点 P 是正方形 $ABCD$ 两条对角线的交点。请问三角形 FPD 的面积是多少？



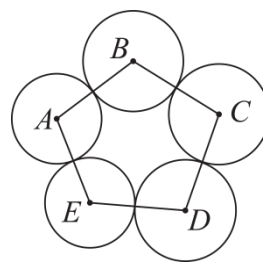
- (A) 14.5 cm^2
 (B) 15 cm^2
 (C) 15.75 cm^2
 (D) 16.5 cm^2
 (E) 17.5 cm^2

28. The positive integer N is such that the product of its digits is 20. Which of the following could not be the product of the digits of $N + 1$?

正整数 N 的各个数位上的数字之积为 20。请问下列哪一项不可能是 $N + 1$ 的各个数位上数字的乘积？

- (A) 40 (B) 30 (C) 25 (D) 35 (E) 24

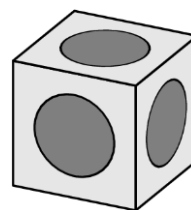
29. Five circles with centres A , B , C , D and E are arranged as shown in the diagram. Line segments are drawn to join the centres of adjacent circles. It is known that $AB = 16$ cm, $BC = 14$ cm, $CD = 17$ cm, $DE = 13$ cm, $AE = 14$ cm. Which point is the centre of the circle with the largest radius?



圆心为 A 、 B 、 C 、 D 和 E 的五个圆的排列方式如图所示。将相邻两个圆的圆心用线段连接起来。已知 $AB = 16$ cm, $BC = 14$ cm, $CD = 17$ cm, $DE = 13$ cm, $AE = 14$ cm。请问以哪个点为圆心的圆的半径最大?

- (A) A (B) B (C) C (D) D (E) E

30. A hole in the shape of a hemisphere is carved into each face of a cube. The holes are identical and centered at the centre of each face. The holes touch their neighbours at only one point. The cube has side 2. What is the diameter of each hole?

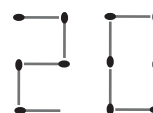


将立方体的每个面都挖掉一个半球形的洞。这些洞完全相同，且球心与每个面的中心重合。这些洞面与它相邻的洞面仅有一个切点。已知该立方体的边长为 2，请问每个洞的直径是多少?

- (A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{3}{2}$ (E) $\sqrt{\frac{3}{2}}$

3 points

1 (EE). Carola is forming the four-digit number 2022 using some matches from a box. The box originally contained 30 matches. She has already started and formed the first two digits, as shown in the diagram. How many matches will remain in the box when she has finished forming 2022?



- (A) 20 (B) 19 (C) 10 (D) 9 (E) 5

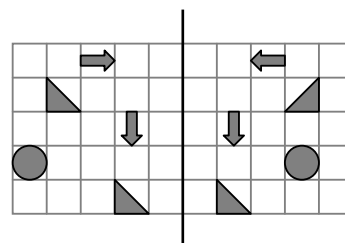
SOLUTION: Carola need 5 matches for a digit 2 and 6 matches for a digit 0. So she need $3 \cdot 5 + 1 \cdot 6 = 21$ matches and $9 = 30 - 21$ matches are left in the box.

2 (FR). An equilateral triangle of side 12 has the same perimeter as a square of side x . What is the value of x ?

- (A) 9 (B) 12 (C) 16 (D) 24 (E) 36

SOLUTION: An equilateral triangle of side 12 has a perimeter of 36. The perimeter of the square is four times the length of the side x and we get $4x = 36$. So the length of the side of the square with the same perimeter is $\frac{36}{4} = 9$.

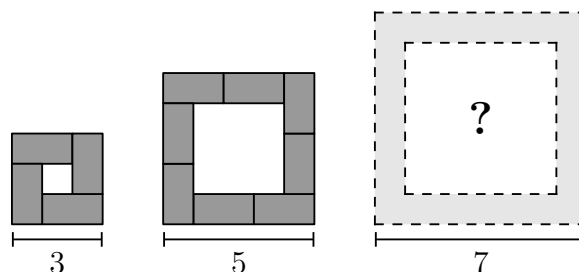
3 (GR). Some shapes are drawn on a piece of paper. The teacher folded the left-hand side of the paper over the thick line. How many of the shapes on the left-hand side will fit exactly on top of a shape on the right-hand side?



- (A) 1 (B) 2 (C) 3
(D) 4 (E) 5

SOLUTION: Two figures will not match - circle because of the distance and triangle in the lower line because of orientation. The other 3 WILL match.

4 (DK). Katrin arranges tables of size 2×1  according to the number of participants in a meeting. The diagrams show a top view of the tables for a small, a medium and a large meeting. How many tables are used for the large meeting?

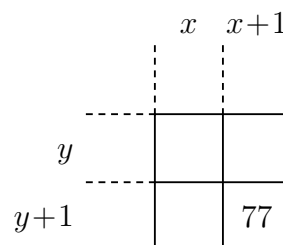


- (A) 10 (B) 11 (C) 12 (D) 14 (E) 16

SOLUTION: Each side has 3 tables not covering ONE corner, so the total number of tables is $4 \cdot 3 = 12$. Alternative: $7^2 - 5^2 = 24$, $24 : 2 = 12$ tables.

5 (FR). A square of numbers is taken out from a multiplication table. Only one number is visible. The integers x and y are both positive and x is greater than y . What is the value of x ?

- (A) 6 (B) 7 (C) 8
(D) 10 (E) 11



SOLUTION: We have $(x + 1)(y + 1) = 77 = 11 \cdot 7$. So $x + 1 = 11 \Rightarrow x = 10$.

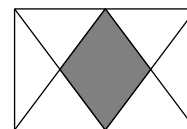
6 (FR). I am less than my half and greater than my double. The sum of me and my square is zero. Who am I?

- (A) -2 (B) -1 (C) 0 (D) 1
(E) 2

SOLUTION: The second condition shows that the number is either 0 or -1 . The first condition excludes 0 so the right answer is B.

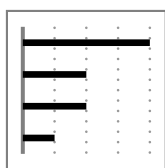
7 (DE). In the rectangle shown, the midpoints of the two longer sides are joined to all four vertices. What fraction of the rectangle is shaded?

- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{2}{7}$ (D) $\frac{1}{3}$ (E) $\frac{2}{5}$

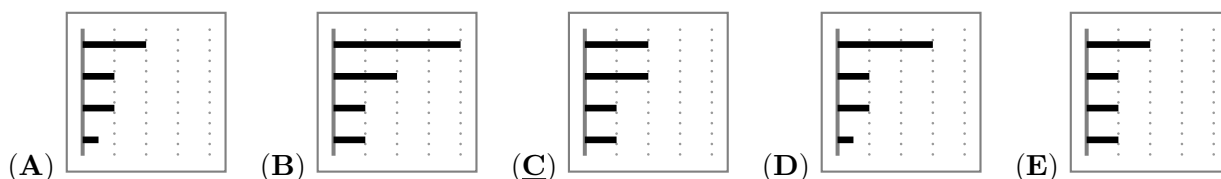


SOLUTION: If you cut the rectangle vertically in the middle, you see that 2 of the 8 triangles are shaded. So $\frac{1}{4}$ of the rectangle is shaded.

8 (DE). On Nadya's smartphone, this diagram shows how much time she spent last week on each of her apps.



This week she halved the time spent on two of these apps, but spent the same amount of time on the other two apps. Which of the following could be the diagram for this week?



SOLUTION: In diagram B the time of only one app was halved, in D and E the times of three apps were halved. In the diagram A all four times have been halved. C is the correct answer.

9 (US). There are five candidates in the school election. After 90% of the votes had been counted, the preliminary results were as follows:

Alex	Bella	Calvin	Diane	Eddy
14	11	10	8	2

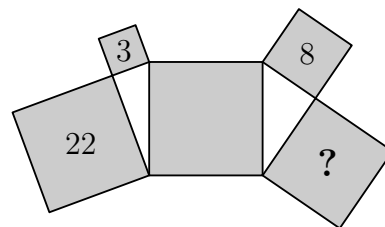
How many students still have a chance of winning the election?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: Answer: (C) Solution: In total, $14 + 11 + 10 + 8 + 2 = 45$ votes have been cast. This is 90% of the total, so 50 votes were cast in total. This means that 5 votes have not been counted yet.

For each candidate, the best case is if they get all of the remaining votes, and nobody else gets any of the votes. In each of those cases: Alex will get 19 votes and win. The votes are 19, 11, 10, 8, 2. Bella will get 16 votes and win. The votes are 14, 16, 10, 8, 2. Calvin will get 15 votes and win. The votes are 14, 11, 15, 8, 2. Diane will get 13 votes, but she won't win. The votes are 14, 11, 10, 13, 2. Eddy will get 7 votes, but she won't win. The votes are 14, 11, 10, 8, 7. The first three can win, but the last two cannot win even in the best case scenario, so they have no chance of winning. Therefore, the answer is (C) 3.

10 (GR). Five squares and two right-angled triangles are arranged as shown. The numbers 3, 8 and 22 inside three of the squares indicate their areas in square metres. What is the area of the square containing the question mark?

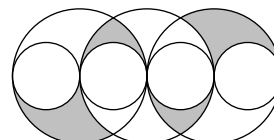


- (A) $14m^2$ (B) $15m^2$ (C) $16m^2$ (D) $17m^2$ (E) $18m^2$

SOLUTION: By Pythagoras the area of the central square is both equal to $3 + 22$ and to $8 + X$. So $X = 17$

4 points

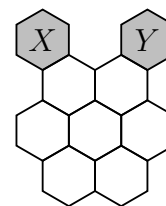
11 (CT). The diagram shows three large circles of equal radius and four small circles of equal radius where the centers of all circles and all points of contact lie on one straight line. The radius of each small circle is 1. What is the shaded area?



- (A) π (B) 2π (C) 3π (D) 4π (E) 6π

SOLUTION: The shaded areas in the right most large circle fit exactly to two of the the white parts of the first large circle from the left. Therefore the shaded area is exactly that of one of the large circles minus the area of two small circles. $A = 4\pi - 2\pi = 2\pi$.

12 (FR). Apini moves from hexagon X to hexagon Y . She can only move from one hexagon to another if they have an edge in common. How many different routes are there from X to Y that pass through each of the seven white hexagons exactly once?



- (A) 2 (B) 3 (C) 4
(D) 5 (E) 6

SOLUTION: Considering each of the 5 hexagons that Apini can pass just after the central hexagon, there is only one possible path for each.

In more detail: If you number the outer hexagons from 1 to 6 counterclockwise then it is clear Apini can visit them only in their respective order and pass the central hexagon only between one pair of neighbouring numbers. There are five pairs of neighbouring numbers, so there are five different possible routes.

13 (AU). I once met six siblings whose ages were six consecutive whole numbers. I asked each of them the question: "How old is your oldest sibling?" Which of the following could **not** be the sum of their six answers?

- (A) 95 (B) 125 (C) 167 (D) 205 (E) 233

SOLUTION: Suppose that the age of the oldest sibling is a . Then the other five siblings all answer a , while the oldest sibling answers $a - 1$. So the sum of their six answers is $5a + (a - 1) = 6a - 1$ and must be one fewer than a multiple of 6. The only number that is not one fewer than a multiple of 6 is 205.

14 (SE). Eva puts 2022 tiles in a long line. Then Adam removes every sixth tile. Next Beata removes every fifth tile from those that remain. Then Calle removes every fourth tile. Finally, Doris removes all the remaining tiles. How many tiles does Doris remove?

- (A) 0 (B) 337 (C) 674 (D) 1011 (E) 1348

SOLUTION: 1: After Adam removes his part of tiles there are $2022 \cdot \frac{5}{6}$ tiles left. Similarly after Beata's move there will be $2022 \cdot \frac{5}{6} \cdot \frac{4}{5}$ and finally after Calle's move we'll have $2022 \cdot \frac{5}{6} \cdot \frac{4}{5} \cdot \frac{3}{4} = 2022 \cdot \frac{3}{6} = 1011$ tiles.

2: each person removes 337 tiles so there are $2022 - 3 \cdot 337 = 1011$ tiles left.

3: (a cute one) split all tiles in groups of 6. Then Adam, Beata und Calle removes exactly the last one, so a half of original tiles is left for Doris.

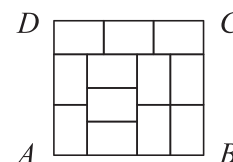
15 (LV). Three children asked their grandmother how old she was. She replied by asking them to guess her age. One child said she was 75, one said she was 78 and one said she was 81. It turned out that one of the guesses was wrong by 1 year, one was wrong by 2 years and one was wrong by 4 years. What is the grandmother's age?

- (A) 76 (B) 77 (C) 79 (D) 80
(E) cannot be determined exactly

SOLUTION: The grandmother could be $77 = 75 + 2 = 78 - 1 = 81 - 4$ or $79 = 75 + 4 = 78 + 1 = 81 - 2$ years old. So one can not determine the age exactly.

16 (RU). The diagram shows a large rectangle $ABCD$ divided into 12 identical small rectangles. What is the ratio AD/DC ?

- (A) $8/9$ (B) $5/6$ (C) $7/8$
(D) $2/3$ (E) $9/8$



SOLUTION: The shorter side of one small rectangle is a , the longer side is b . Then $|DC| = 3b = 3a + b$, so $b = 1.5a$. We look for $\frac{AD}{DC} = \frac{2b+a}{3b} = \frac{4a}{4.5a} = \frac{8}{9}$.

17 (MD). A rabbit and a hedgehog had a race around a 550 m long circular track. Both ran at constant speeds. The rabbit's speed was 10 m/s, and the hedgehog's speed was 1 m/s. They started at the same time. However, the hedgehog ran in the opposite direction to the rabbit. When they met, the hedgehog immediately turned round and ran after the rabbit. How long after the rabbit did the hedgehog reach the finish?

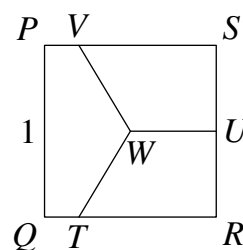
- (A) 45 sec (B) 50 sec (C) 55 sec (D) 100 sec (E) 505 sec

SOLUTION: The rabbit covered the entire distance in $550/10 = 55$ sec. Let x be the distance (m) covered by the hedgehog before they meet. Thus, $x/1 = (550 - x)/10$. It follows that $x = 50$ m. The running time of the hedgehog was $2x/1 = 100$ sec. The hedgehog reached the finish $100 - 55 = 45$ sec after the rabbit.

18 (UK). The diagram shows square $PQRS$ of side-length 1. The midpoint of RS is marked U and the centre of the square is marked W . Line segments TW, UW and VW split the square into three regions of equal area.

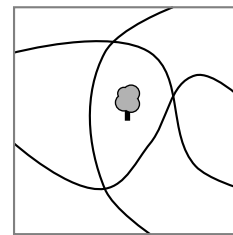
What is the length of SV ?

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$ (E) $\frac{5}{6}$



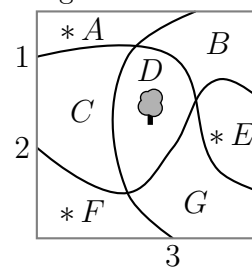
SOLUTION: Since W is the centre of the square, the length of UW is $\frac{1}{2}$. As the square is split into three regions of equal area and the side-length of the square is 1, the area of each region is $\frac{1}{3}$. Let the length of SV be y . Hence $\frac{1}{2} \times \left(y + \frac{1}{2}\right) \times \frac{1}{2} = \frac{1}{3}$. Therefore $y + \frac{1}{2} = \frac{4}{3}$ and hence $y = \frac{5}{6}$.

19 (DE). There are three paths through our city park. A tree is planted in the middle of the park, as shown. What is the smallest number of trees that need to be planted so that there are the same number of trees on both sides of each of the paths?



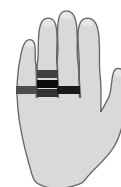
- (A) 1 (B) 2 (C) 3
(D) 4 (E) 5

SOLUTION: Give the regions bounded by the paths as in the picture. Looking at path 1 at least one tree must be planted in one of the regions A, B or E. Looking at path 2 at least one tree must be planted in one of the regions F, G or E. Looking at path 3 at least one tree must be planted in one of the regions A, C and F. But the three mentioned triples above have no region in common. So at least 3 trees must be planted (since the total number of trees must be even). Planting trees at the 3 stars



in the picture for every path the number of trees on both sides is the same.

20 (DE). Veronica has five rings on her fingers, as shown in the diagram. She takes them off one at a time. In how many different ways can she do this?



- (A) 16 (B) 20 (C) 24
(D) 30 (E) 45

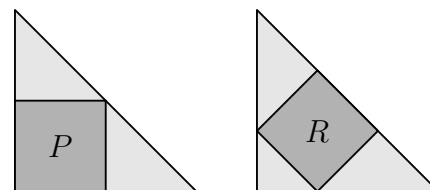
SOLUTION: Veronica must take the 3 rings on the ring finger off in one specific order, let's call the rings A , B , C . She can take off the ring on her pinky before A , between A and B , between B and C or after C , that makes 4 cases. In each of these 4 cases, she can take off the ring on her middle finger on 5 different positions relative to the 4 rings on her pinky or ring finger. That makes $4 \cdot 5 = 20$ different ways.

Second solution: The 3 rings on the ring finger must be taken off in one specific order. The other 2 rings can be taken off at any time. So the number of ways is $5!/3! = 5 \cdot 4 = 20$.

5 points

21 (PR). Two congruent isosceles right-angled triangles each have a square inscribed, as shown in the diagram. The square marked P has an area of 45. What is the area of the square marked R ?

- (A) 35 (B) 40 (C) 45 (D) 50 (E) 60



SOLUTION: Let's split square P in two triangles by its diagonal and square R in four triangles by two diagonals. After splitting square P the larger triangle will consist of 4 small triangles, so the area of the triangle will be equal to 90 units. After splitting square R the larger triangle will consist of 9 small triangles (two larger triangles can be split in two smaller with their height), so the area of the small triangle will be equal to $90 : 9 = 10$ and the area of R will be equal to 40.

22 (CT). Eight teams participate in a football tournament. Each team plays against each other team exactly once. In each match, the winner gets 3 points and the loser does not get any points. If a match is drawn, each team gets 1 point. At the end of the tournament the total number of points obtained by all the teams is 61. What is the largest number of points that the champion team could

have obtained?

- (A) 21 (B) 19 (C) 18 (D) 17 (E) 16

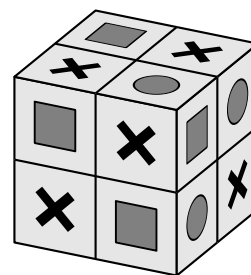
SOLUTION: The minimum sum of points for all teams is $28 \text{ matches} \times 2 \text{ points/match} = 56 \text{ points}$, which occurs in the event that there is a tie in all matches. Each match in which there is a winning team adds one point to the total sum. Thus, $61 - 56 = 5$ points above the minimum indicates that only in 5 games there has been a winning team and in all of them it could have been the champion, which would have obtained $5 \times 3 + 2 \times 1 = 17$ points.

23 (GR). A group of pirates divided 200 gold coins and 600 silver coins between them. Each officer received 5 gold and 10 silver coins. Each sailor received 3 gold and 8 silver coins. Each cabin boy received 1 gold and 6 silver coins. How many pirates are there in the group?

- (A) 50 (B) 60 (C) 72 (D) 80 (E) 90

SOLUTION: We do not try to solve this using lengthy calculations or trial and error, but instead, we use a simple observation. Namely, each rank (officers, sailors, cabin boys) received 5 more silver coins than gold. So if there are N pirates, the excess of silver to gold satisfies $5N = 600 - 200$, so $N = 80$.

24 (GR). The squares on the surface of a $2 \times 2 \times 2$ cube have one of three shapes on them. The shapes are either a circle or a square or an X sign. Any two squares that share a common side have different shapes on them. The picture shows one such possibility. Which of the following combinations of shapes is also possible on such a cube?



- (A) 6 circles, 8 squares and the rest are X's
 (B) 7 circles, 8 squares and the rest are X's
 (C) 5 circles, 8 squares and the rest are X's
 (D) 7 circles, 7 squares and the rest are X's
 (E) none of the previous

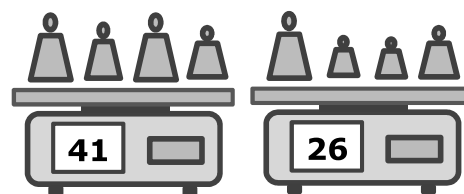
SOLUTION: We do NOT start thinking in a three dimensional space. The beauty of this problem is the simplicity of its solution, based on a simple but nice observation. We show that the only possible combination of shapes is 8 each: Notice that at each vertex three squares share common sides in pairs, so these three squares must each have shape different from the other two. So one of each shape is compulsory at each vertex. Doing this for all 8 vertices we conclude that each shape must appear 8 times. It is easy to construct such a cube. The one on the picture can easily be continued to have the required property.

25 (CT). The inhabitants of a city always speak by means of questions. There are two types of inhabitants: the "positives", who always ask questions for which the answer is "yes" and the "negatives" who always ask questions for which the answer is "no". I met Albert and Berta and Berta asked me "Are Albert and I both negative?". What type of inhabitants are Albert and Berta?

- (A) Both are positives (B) Both are negatives
 (C) Albert positive, Berta negative (D) Albert negative, Berta positive
 (E) There are not enough information to decide.

SOLUTION: It is clear that Berta cannot be positive because the answer would lead to a contradiction. And so, Berta is negative. But if the answer is "No" and we know that she is negative then there cannot be two negatives. Then Albert must be positive.

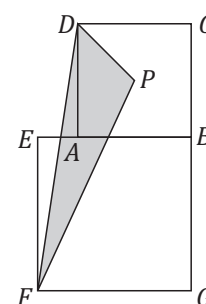
26 (GR). A grocer has twelve different integer weights from 1 kg to 12 kg. She splits them into three groups of four weights each. The total weight of the first group is 41 kg and of the second is 26 kg. Which of the following weights is in the same group as the weight of 9 kg?



- (A) 3 kg (B) 5 kg (C) 7 kg (D) 8 kg (E) 10 kg

SOLUTION: We do not work by trial and error, but use some observations: The total weight is $1 + 2 + \dots + 12 = 78$ kg, so the third group is $78 - 41 - 26 = 11$ kg. Note that as the four smallest weights add up to $1 + 2 + 3 + 4 = 10$ kg, we easily conclude that the only way four weights add up to 11 is the case $1 + 2 + 3 + 5 = 11$. Also, note as the four heaviest weights add up to $12 + 11 + 10 + 9 = 42$, we easily conclude that the only way four weights add up to 41 is the case $12 + 11 + 10 + 8 = 41$. So the three groups are (1, 2, 3, 5), (8, 10, 11, 12) and the rest (4, 6, 7, 9). So from the given answers, only 7 is together with the weight of 9 kg.

27 (EE). The lengths of the diagonals of the squares $ABCD$ and $EFGB$ are 7 cm and 10 cm respectively. The point P is the intersection of the diagonals of the square $ABCD$. What is the area of the triangle FPD ?



- (A) 14.5 cm^2
 (B) 15 cm^2
 (C) 15.75 cm^2
 (D) 16.5 cm^2
 (E) 17.5 cm^2

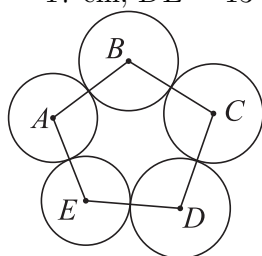
SOLUTION: The triangle FBD is right-angle, as FB and BD are diagonal. Its area is equal to $\frac{1}{2} \cdot 7 \cdot 10 = 35 \text{ cm}^2$. The point P is a midpoint of BD , so FP is median and hence $S_{FPD} = \frac{1}{2} S_{FBD} = 17.5 \text{ cm}^2$.

28 (PE). The positive integer N is such that the product of its digits is 20. Which of the following could not be the product of the digits of $N + 1$?

- (A) 40 (B) 30 (C) 25 (D) 35 (E) 24

SOLUTION: N could consist only of digits 1, 2, 4, 5. Therefore $N + 1$ couldn't have the digit 7 in it, so the product of its digit can't be equal to 35. 40 can be obtained, when $N = 2251$ and $N + 1 = 2252$. 30 can be obtained, when $N = 2512$ and $N + 1 = 2513$. 25 can be obtained, when $N = 54$ and $N + 1 = 55$. 24 can be obtained, when $N = 45$ and $N + 1 = 46$.

29 (RU). Five circles with centres A , B , C , D and E are arranged as shown in the diagram. Line segments are drawn to join the centres of adjacent circles. It is known that $AB = 16$ cm, $BC = 14$ cm, $CD = 17$ cm, $DE = 13$ cm, $AE = 14$ cm. Which point is the centre of the circle with the largest radius?



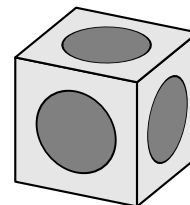
radius?

- (A) A (B) B (C) C (D) D (E) E

SOLUTION: Suppose that r_a, r_b, r_c, r_d, r_e are radiuses of the circles in the centers at point A, B, C, D, E respectively. Then $r_a + r_b = 16$ cm, $r_b + r_c = 14$ cm, $r_c + r_d = 17$ cm, $r_d + r_e = 13$ cm, $r_e + r_a = 14$

cm. Therefore $2(r_a + r_b + r_c + r_d + r_e) = 74$ and $r_a + r_b + r_c + r_d + r_e = 37$. Hence $r_a = (r_a + r_b + r_c + r_d + r_e) - (r_b + r_c) - (r_d + r_e) = 10$ cm is the radius of the largest circle with center in A (as BC and DE are the smallest of five).

30 (FI). A hole in the shape of a hemisphere is carved into each face of a cube. The holes are identical and centered at the centre of each face. The holes touch their neighbours at only one point. The cube has side 2. What is the diameter of each hole?



- (A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{3}{2}$ (E) $\sqrt{\frac{3}{2}}$

SOLUTION: The problem can be reduced to 2D by cutting the cube in halves by a horizontal plane passing through the center. In this plane the holes are seen as semicircles tangent to each other. By Pythagoras we have $2r = \sqrt{1^2 + 1^2} = \sqrt{2}$.